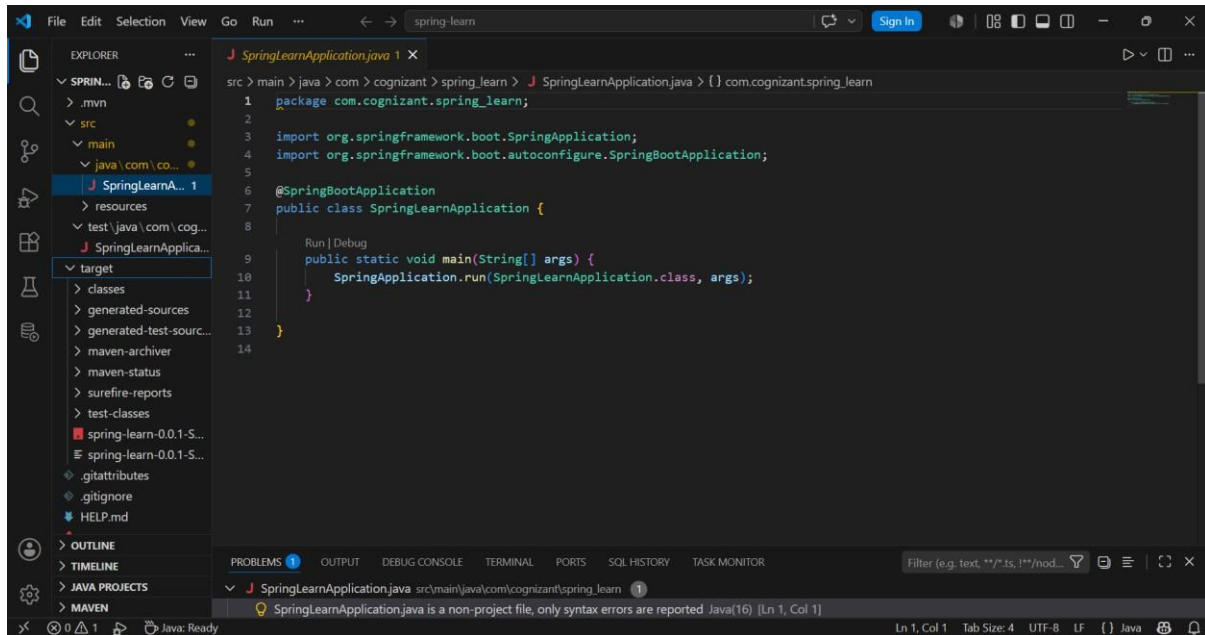
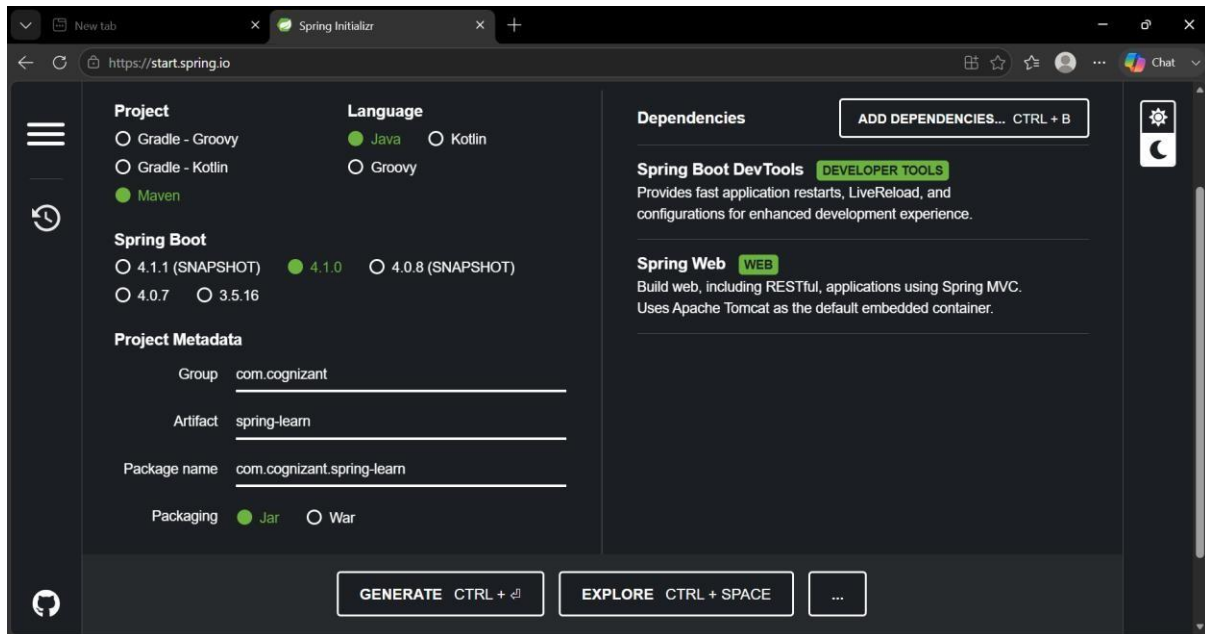


DIGITAL NURTURE 5.0 DEEPSKILLING

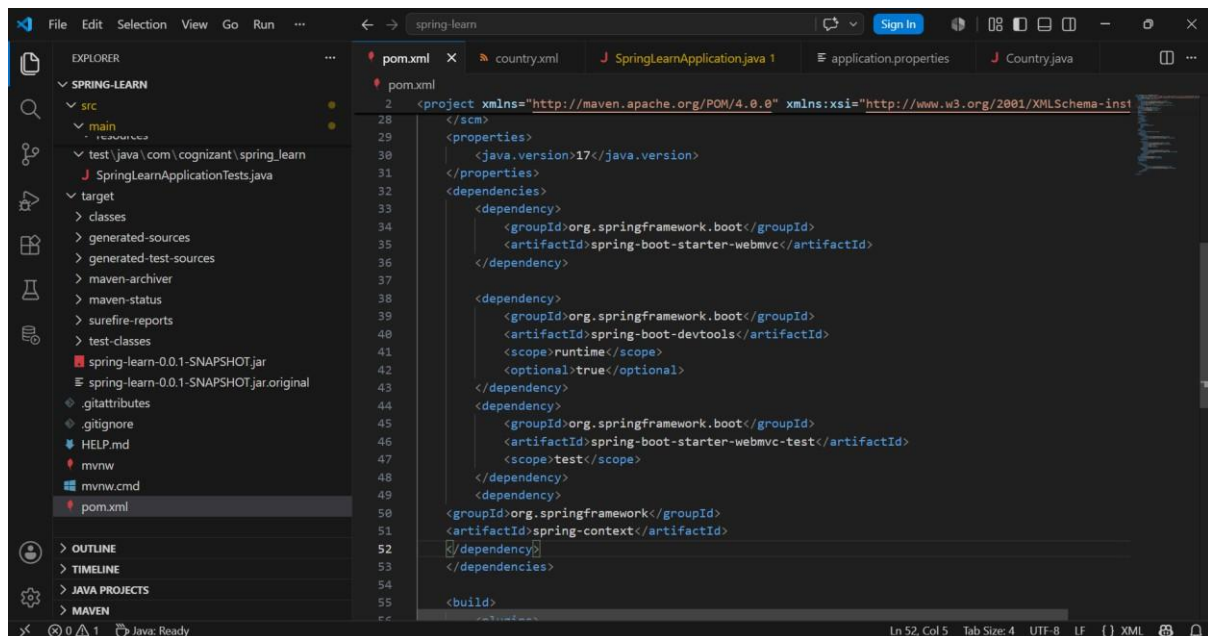
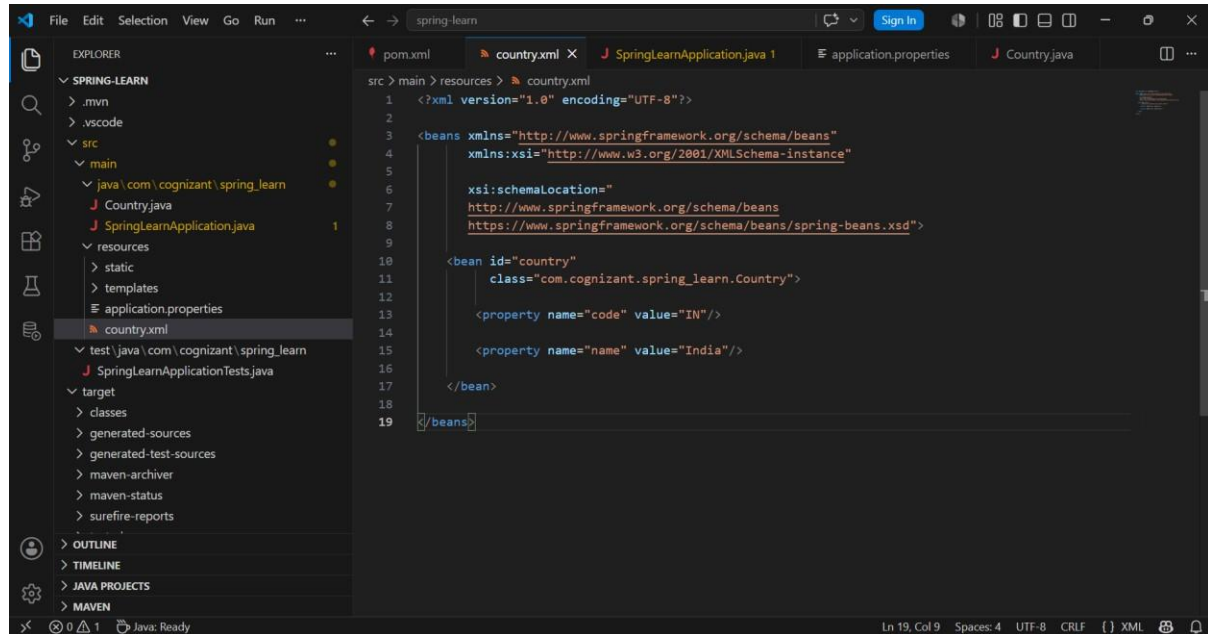
WEEK 3 – HANDS-ON

Spring REST using Spring Boot 3

Create a Spring Web Project using Maven

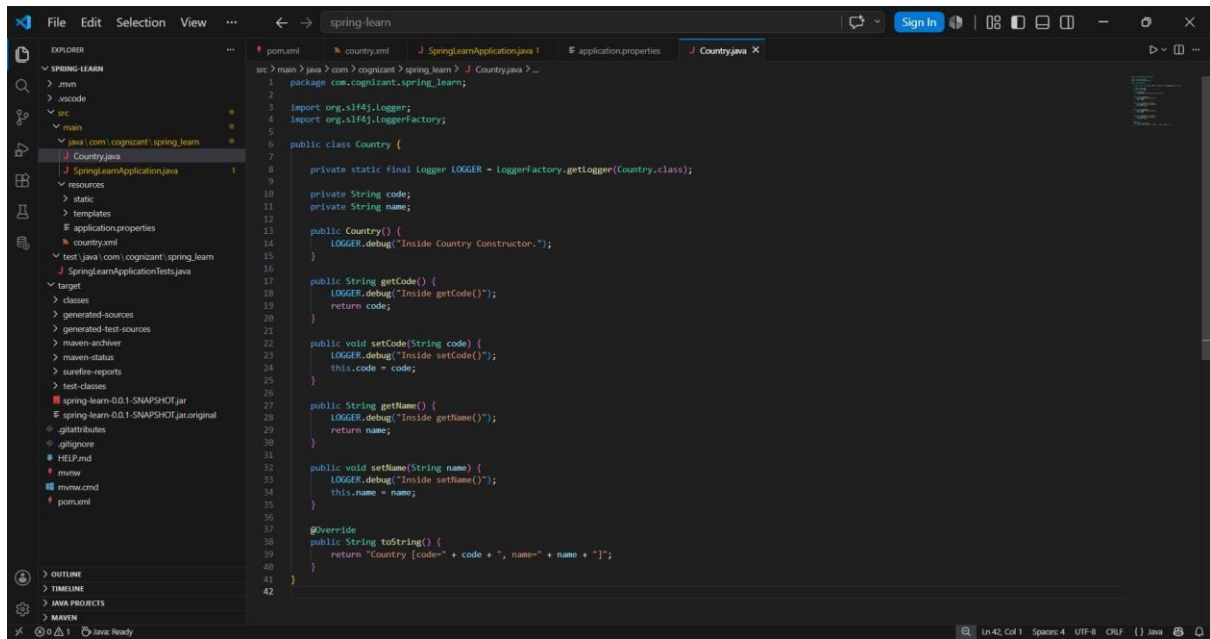


Spring Core – Load Country from Spring Configuration XML

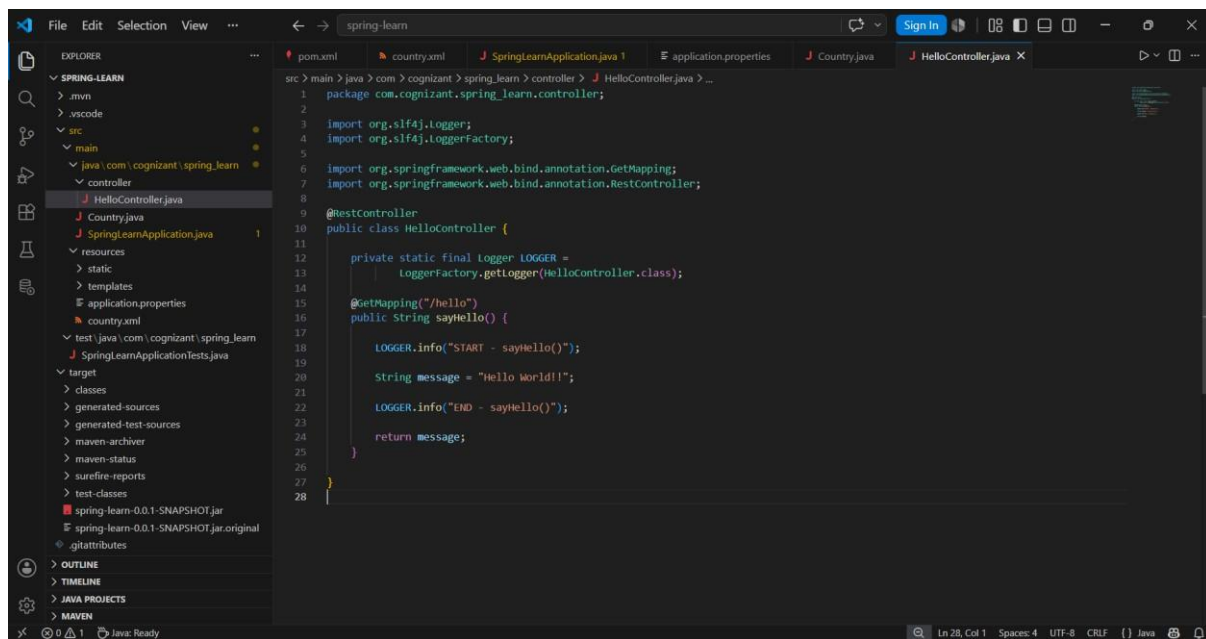


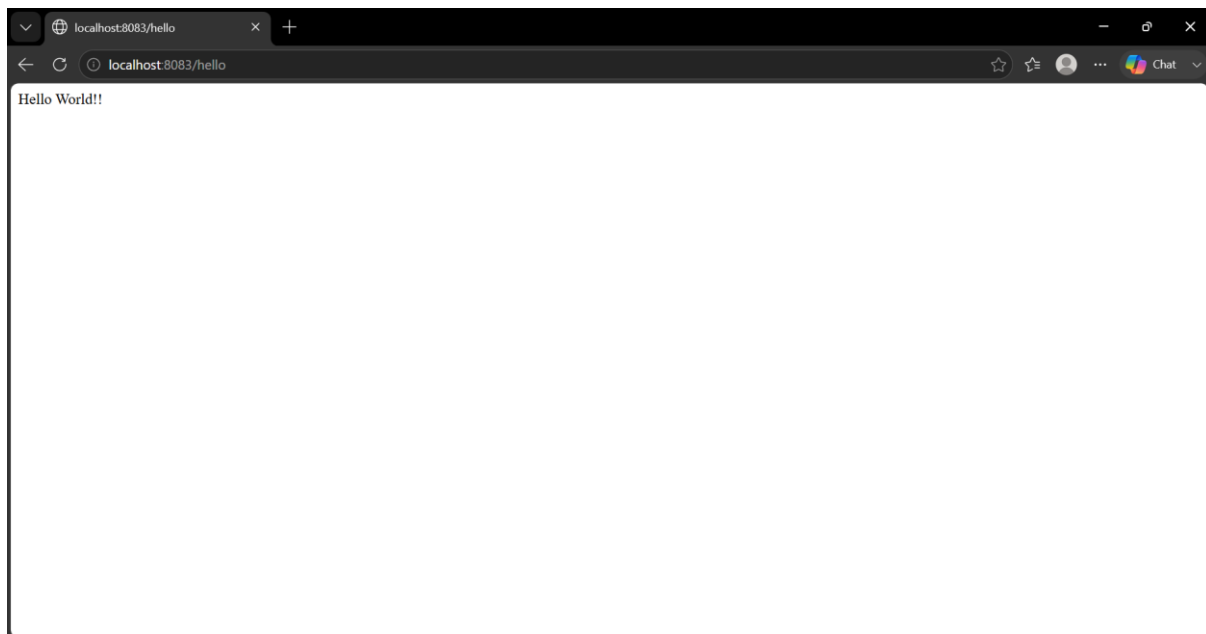
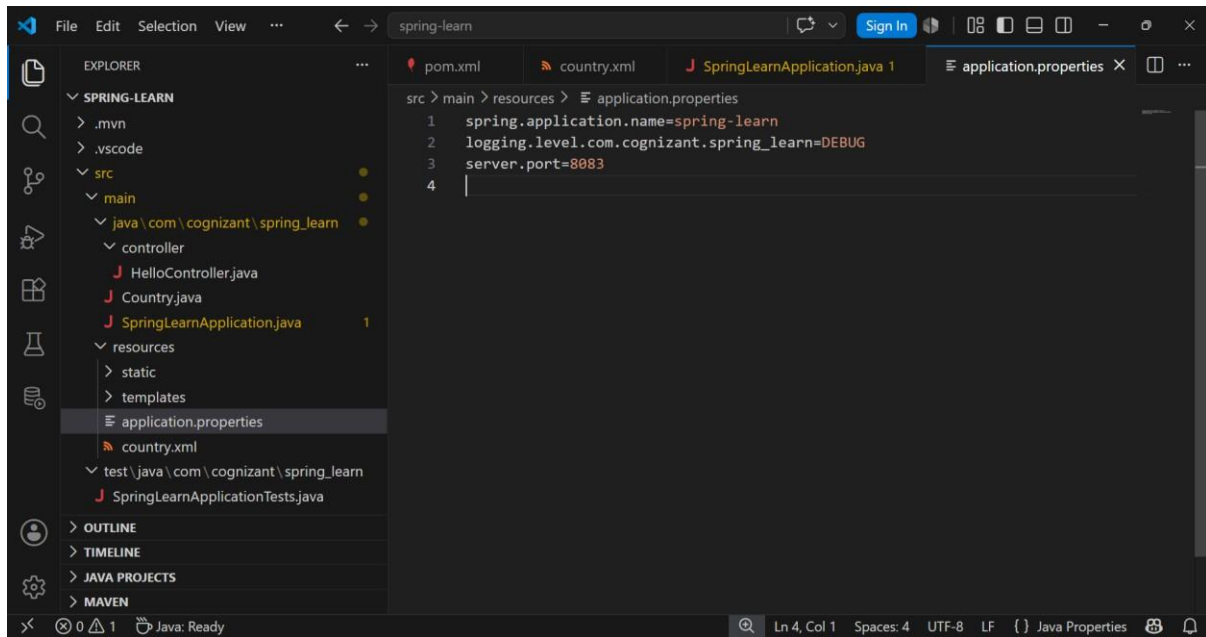
```
src > main > java > com > cognizant > spring_learn > J SpringLearnApplication.java > % SpringLearnApplication
1 package com.cognizant.spring_learn;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 import org.springframework.context.ApplicationContext;
7 import org.springframework.context.support.ClassPathXmlApplicationContext;
8
9 import org.springframework.boot.SpringApplication;
10 import org.springframework.boot.autoconfigure.SpringBootApplication;
11
12 @SpringBootApplication
13 public class SpringLearnApplication {
14
15     // Logger declaration
16     private static final Logger LOGGER = LoggerFactory.getLogger(SpringLearnApplication.class);
17
18     Run[Debug]
19     public static void main(String[] args) {
20
21         SpringApplication.run(SpringLearnApplication.class, args);
22
23         // Call the method to load country bean
24         displayCountry();
25     }
26
27     public static void displayCountry() {
28
29         // Load the Spring XML configuration
30         ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
31
32         // Get the country bean
33         Country country = context.getBean("country", Country.class);
34
35         // Display country details
36         LOGGER.debug("Country : {}", country.toString());
37     }
38 }
```

```
src > main > resources > application.properties
1 spring.application.name=spring_learn
2 logging.level.com.cognizant.spring_learn=DEBUG
```

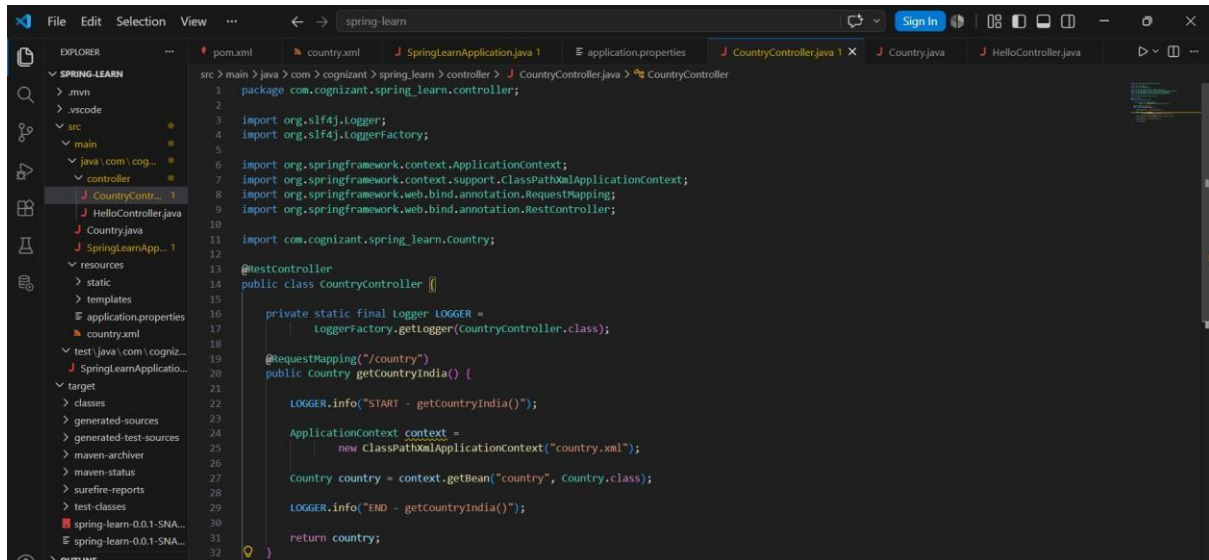


Hello World RESTful Web Service



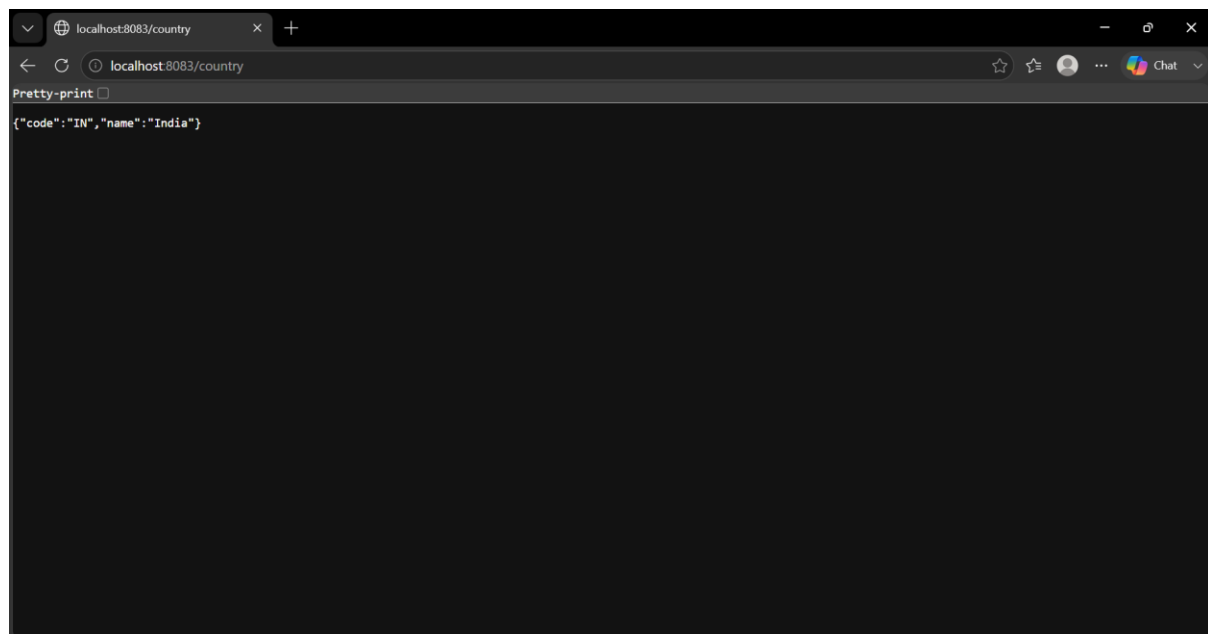


REST - Country Web Service



The screenshot shows a code editor with the following Java code for `CountryController.java`:

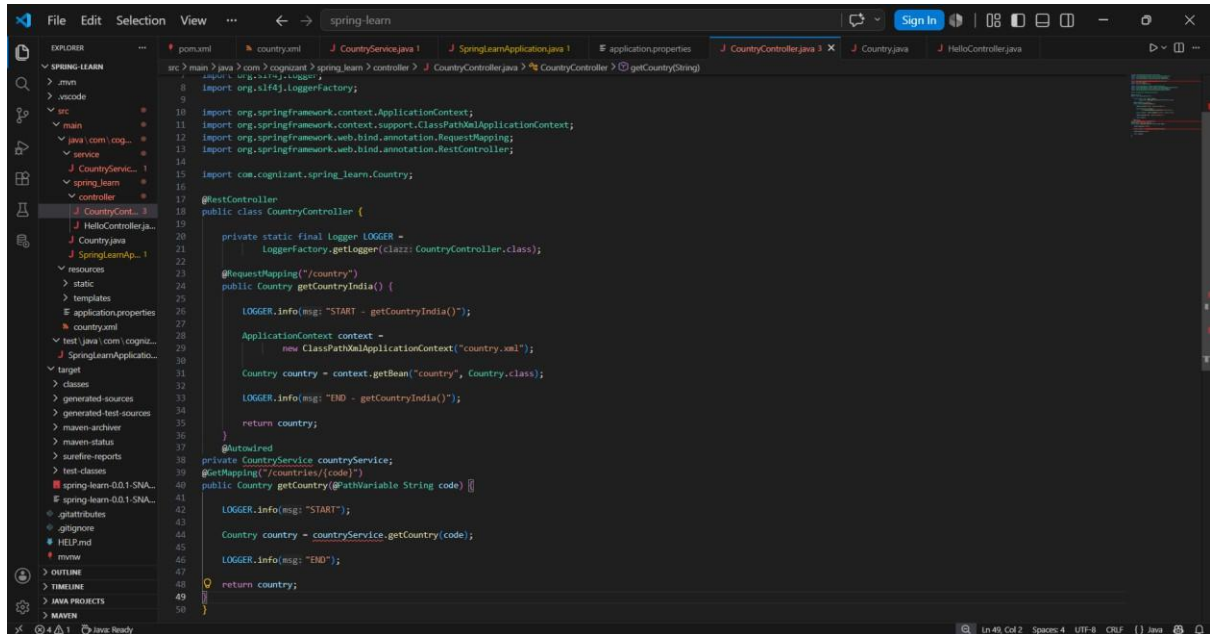
```
1 package com.cognizant.spring_learn.controller;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 import org.springframework.context.ApplicationContext;
7 import org.springframework.context.support.ClassPathXmlApplicationContext;
8 import org.springframework.web.bind.annotation.RequestMapping;
9 import org.springframework.web.bind.annotation.RestController;
10
11 import com.cognizant.spring_learn.Country;
12
13 @RestController
14 public class CountryController {
15
16     private static final Logger LOGGER =
17         LoggerFactory.getLogger(CountryController.class);
18
19     @RequestMapping("/country")
20     public Country getCountryIndia() {
21
22         LOGGER.info("START - getCountryIndia()");
23
24         ApplicationContext context =
25             new ClassPathXmlApplicationContext("country.xml");
26
27         Country country = context.getBean("country", Country.class);
28
29         LOGGER.info("END - getCountryIndia()");
30
31         return country;
32     }
33 }
```



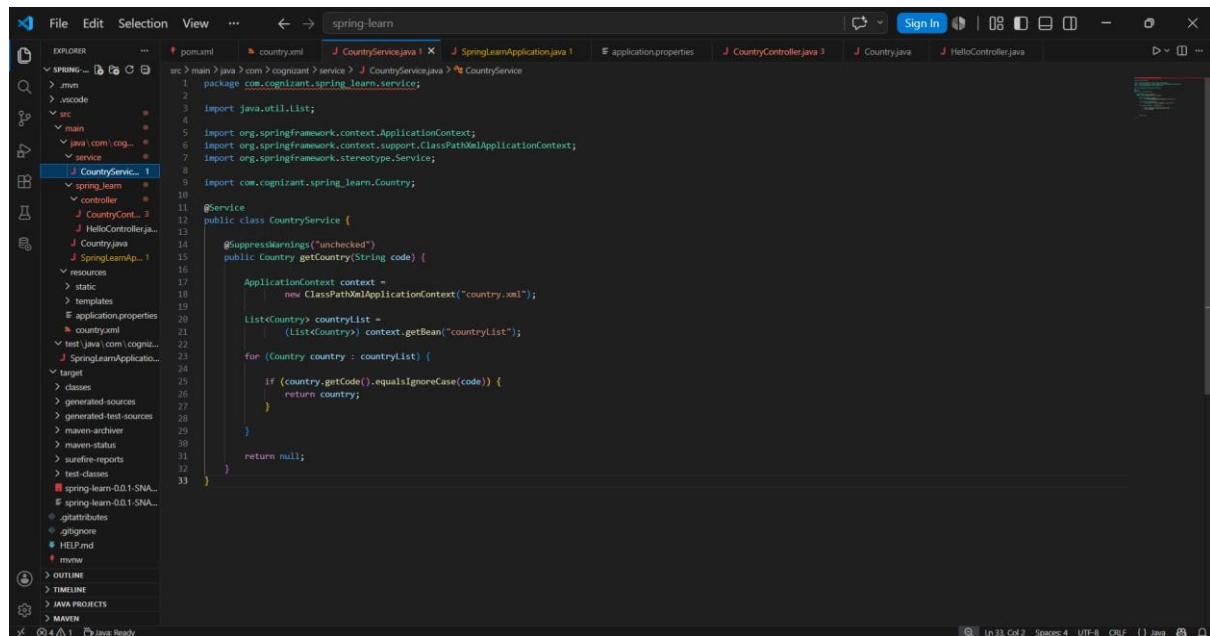
The screenshot shows a web browser window with the address bar displaying `localhost:8083/country`. The page content shows the JSON response of the API call:

```
{"code":"IN","name":"India"}
```

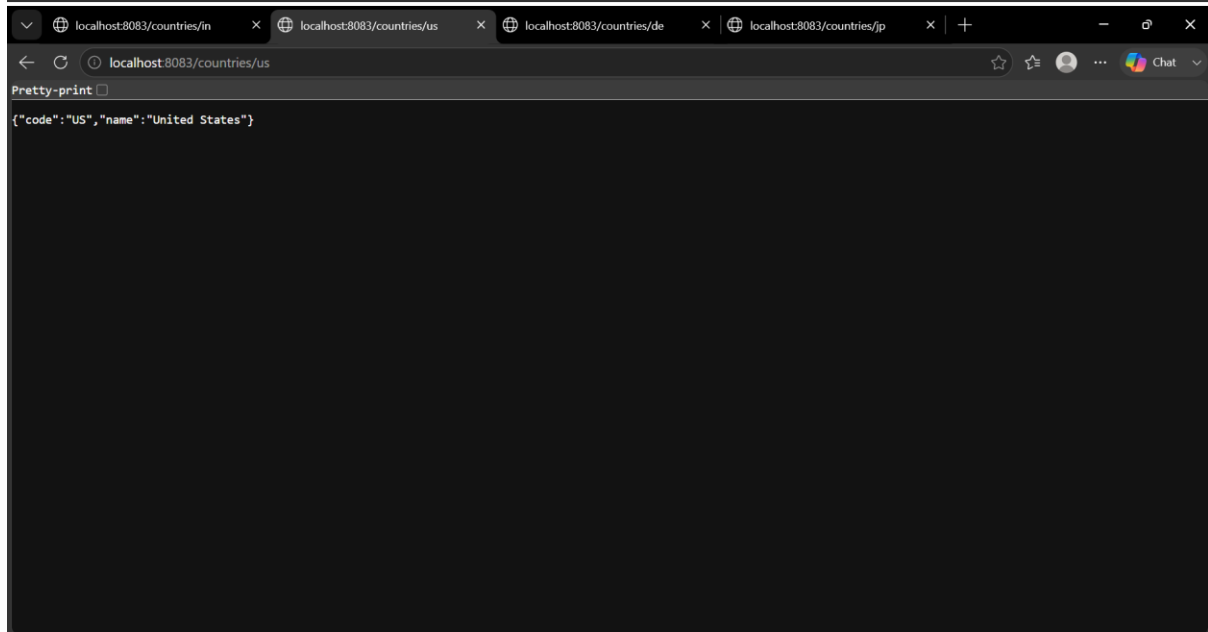
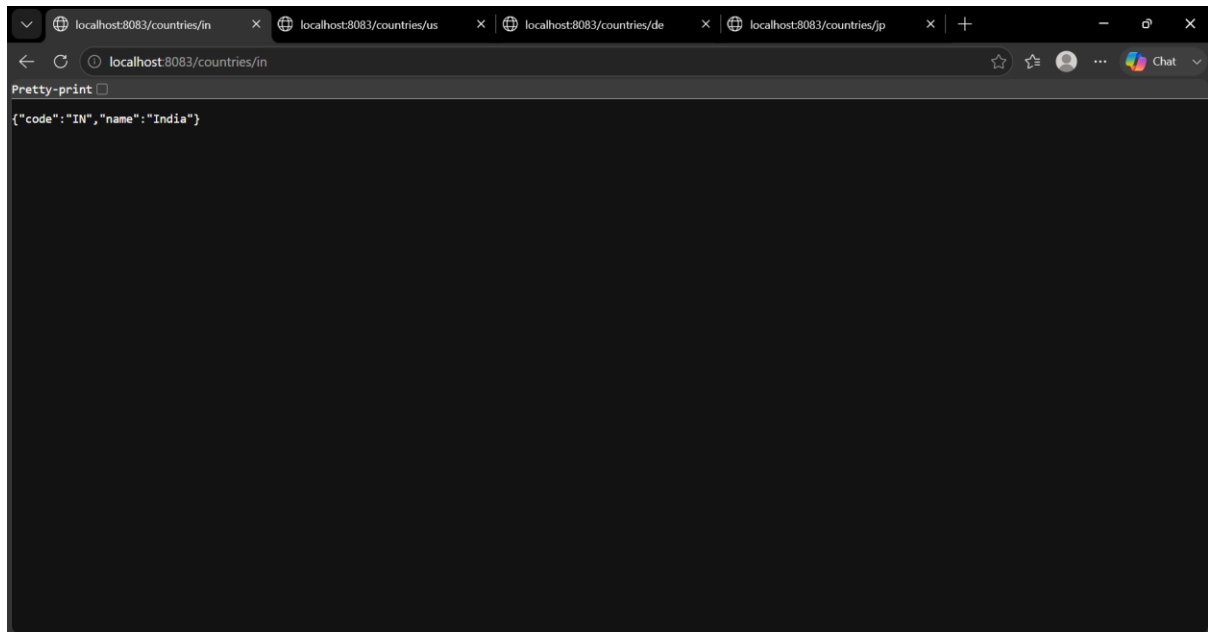

REST - Get country based on country code

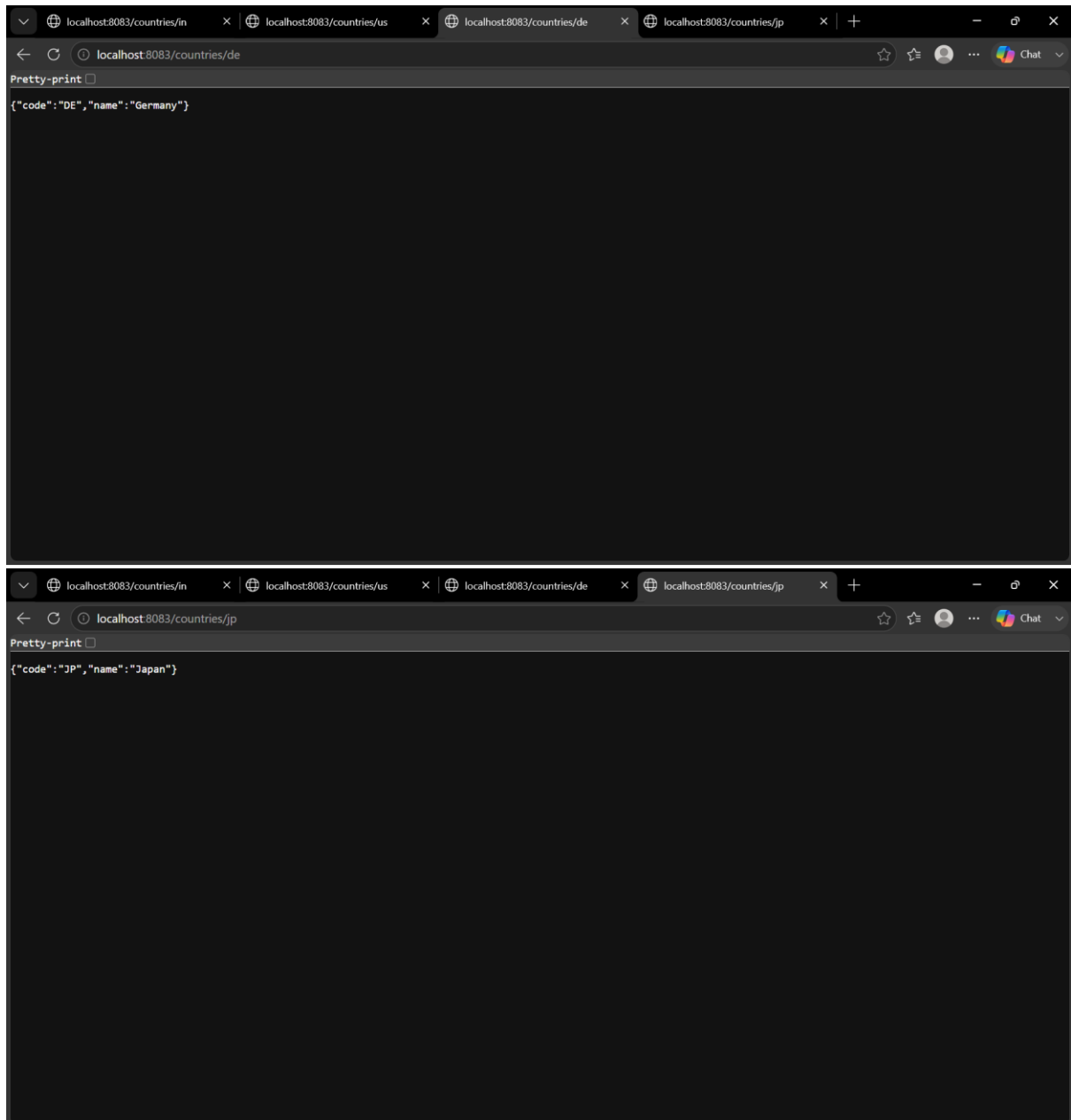


```
src > main > java > com > cognizant > spring_learn > controller > J CountryController.java > % CountryController > @getCountry(String)
1  import org.slf4j.Logger;
2  import org.slf4j.LoggerFactory;
3
4  import org.springframework.context.ApplicationContext;
5  import org.springframework.context.support.ClassPathXmlApplicationContext;
6  import org.springframework.web.bind.annotation.RequestMapping;
7  import org.springframework.web.bind.annotation.RestController;
8
9  import com.cognizant.spring_learn.Country;
10
11  @RestController
12  public class CountryController {
13
14      private static final Logger LOGGER =
15          LoggerFactory.getLogger(ClassPathXmlApplicationContext.class);
16
17      @RequestMapping("/country")
18      public Country getCountryIndia() {
19
20          LOGGER.info(msg: "START - getCountryIndia()");
21
22          ApplicationContext context =
23              new ClassPathXmlApplicationContext("country.xml");
24
25          Country country = context.getBean("country", Country.class);
26
27          LOGGER.info(msg: "END - getCountryIndia()");
28
29          return country;
30      }
31
32      @Autowired
33      private CountryService countryService;
34      @RequestMapping("/countries/{code}")
35      public Country getCountry(@PathVariable String code) {
36
37          LOGGER.info(msg: "START");
38
39          Country country = countryService.getCountry(code);
40
41          LOGGER.info(msg: "END");
42
43          return country;
44      }
45  }
```



```
src > main > java > com > cognizant > service > J CountryService.java > % CountryService
1  package com.cognizant.spring_learn.service;
2
3  import java.util.List;
4
5  import org.springframework.context.ApplicationContext;
6  import org.springframework.context.support.ClassPathXmlApplicationContext;
7  import org.springframework.stereotype.Service;
8
9  import com.cognizant.spring_learn.Country;
10
11  @Service
12  public class CountryService {
13
14      @SuppressWarnings("unchecked")
15      public Country getCountry(String code) {
16
17          ApplicationContext context =
18              new ClassPathXmlApplicationContext("country.xml");
19
20          List<Country> countryList =
21              (List<Country>) context.getBean("countryList");
22
23          for (Country country : countryList) {
24
25              if (country.getCode().equalsIgnoreCase(code)) {
26                  return country;
27              }
28          }
29
30          return null;
31      }
32  }
```





Create authentication service that returns JWT

